

$$X_d = \sqrt{(Z_{sd}^2 - R_a^2)} = \underline{\hspace{2cm}} \Omega$$

$$X_q = \sqrt{(Z_{sq}^2 - R_a^2)} = \underline{\hspace{2cm}} \Omega$$

a) To find percentage regulation at full load and 0.8 pf lag

$$V = 230V, I = \underline{\hspace{2cm}} A, \phi = +36.87^\circ$$

$$\Psi = \tan^{-1} \left(\frac{V \sin \phi + IX_q}{V \cos \phi + IR_a} \right) = \underline{\hspace{2cm}}^\circ$$

$$\delta = \Psi - \phi = \underline{\hspace{2cm}}^\circ$$

$$E_f = V \cos \delta + IR_a \cos \Psi + IX_d \sin \Psi = \underline{\hspace{2cm}} V$$

$$\% \text{ regulation} = \frac{E_f - V}{V} \times 100 \% = \underline{\hspace{2cm}} \%$$

% regulation at full load : $V = \underline{\hspace{2cm}} V, I = \underline{\hspace{2cm}} A$

Power factor	ϕ	ψ	δ	E_f	Regulation
0 lag					
0.2 lag					
0.4 lag					
0.6 lag					
0.8 lag					
1					
0.8 lead					
0.6 lead					
0.4 lead					
0.2 lead					
0 lead					

b) To find percentage regulation at full load and 0.8 pf lead

$$V = 230V, I = \underline{\hspace{2cm}} A, \phi = -36.87^\circ$$

$$\Psi = \tan^{-1} \left(\frac{V \sin \phi + IX_q}{V \cos \phi + IR_a} \right) = \underline{\hspace{2cm}}^\circ$$